

CARSON FULMER

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EDUCATION

University of Florida

Bachelor of Science in Computer Science *GPA: 3.5*

Relevant Coursework: Data Structures, Object-Oriented Programming, Computer Organization

Gainesville, FL

January 2024 - May 2026

SKILLS

Languages & Frameworks: C++, C, TypeScript, Python, JavaScript, React, QT Creator, Express
Tools & Technologies: Git, GitHub, MongoDB, RESTful APIs, Google Cloud, Linux, JWT Auth, Pandas
Core Competencies: Algorithms, Data Structures, Concurrent Programming, Cloud Deployment, UI/UX Design, Database Management, API Integration, Full-Stack Development

PROFESSIONAL EXPERIENCE

University of Florida - Computer Science Department

Undergraduate Research Assistant

Gainesville, FL

January 2025 - Present

- Engineered a distributed simulation platform using Ponylang to accelerate agent-based simulations
- Designed scalable architecture supporting multiple concurrent users and collaborated with faculty to improve computational efficiency of the platform and provide oversight as well as feedback
- Applied actor-based concurrency patterns to create fault-tolerant systems capable of handling complex scientific simulations preventing deadlocks and variable races, allowing for a safe concurrent program

Intercruises - Shoreside & Port Services

Customer Service Representative

Tampa, FL

August 2021 - December 2023

- Facilitated experiences for thousands of monthly travelers while resolving numerous customer inquiries with consistently positive feedback and communication between other employees
- Coordinated with customs officials to optimize passenger processing and improve terminal transit times
- Developed knowledge of travel regulations and utilized effective communication skills in high-pressure environments

SOFTWARE DEVELOPMENT PROJECTS

Global Health Analysis - TypeScript, C++, Chart.js, React

github.com/carsonful/World-Disease-Analysis

A data analytics platform providing interactive visualizations of global health trends and disease patterns.

- Engineered a C++ backend system that significantly improved data processing speed for complex global health datasets which included over 1+ million entries that were optimized for query
- Created responsive UI supporting data for 150+ countries with high browser compatibility and seamless JSON integration allowing for easy readability of data to display for users with filtering of graphs
- Implemented dynamic filtering capabilities allowing users to compare health metrics across regions and time periods

Starcrosser - Visualization Tool - C++, QT Creator

github.com/fmcubium/Starcrosser

An educational tool that helps users comprehend vast interstellar distances using familiar vehicle references.

- Constructed an interactive visualization application with optimized path-finding algorithms for efficient route calculations among different stars in our universe using a dataset of over 100,000 stars
- Designed memory-efficient data structures and led Git-based development resulting in minimal bugs at release
- Created intuitive UI allowing users to compare travel times between star systems using familiar vehicle references, in which some algorithms provided 60% faster calculations between stars

DoneTogether - Goal Platform - TypeScript, React, MongoDB, Express, Authentication

github.com/oixel/DoneTogether

A collaborative goal-tracking platform connecting users with similar aspirations for mutual accountability and support.

- Created a full-stack web application that increases user accountability through a collaborative goal-tracking approach where users can invite their peers to a shared goal which is shown to improve completion rate
- Architected RESTful API endpoints with React UI components to deliver a responsive, engaging user experience while continuing to be easy to navigate and/or sign up for new users
- Integrated social features like progress sharing, milestone celebrations, and collaborative challenges to enhance motivation toward completing the shared goal between users and their peers

Canvas Calendar Notification System - Python, APIs, Google Cloud

github.com/carsonful/canvascalendar

An automated system that fetches Canvas LMS assignments and delivers personalized notifications with weather updates.

- Developed a notification system that improved assignment completion rates among University of Florida students
- Leveraged Pandas for reliable data processing and deployed the solution on Google Cloud with high availability
- Implemented intelligent priority sorting to highlight urgent assignments and provide customizable notification scheduling using a builtin function in the virtual machine called a cronjob

CERTIFICATIONS

Microsoft Technology Associate (MTA): Python Programming, Security Fundamentals

April 2022